

Sensor-Based Smart Street Lighting System Using Arduino and LDR

1. Objective

To design and simulate an Arduino-based smart street lighting system that uses a Light Dependent Resistor (LDR) to detect ambient light. The LEDs automatically operate when darkness is detected, demonstrating an energy-efficient lighting solution.

2. Components Used

Component	Quantity
Arduino Uno	1
Photoresistor (LDR)	1
LEDs	5
220 Ω Resistors	5
10 k Ω Resistor	1
Breadboard	1
Jumper Wires	As required

3. Working Principle

The photoresistor senses the surrounding light intensity. The Arduino continuously reads the analog value from the LDR through pin A0. When the light intensity falls below a predefined threshold (indicating darkness), the Arduino activates the LEDs in a running sequence. When sufficient light is detected, the LEDs remain OFF, helping conserve energy.

4. Code:

```
int ldr = A0;
void setup()
{
    for(int i = 1; i <= 5; i++)
        pinMode(i, OUTPUT);

    Serial.begin(9600);
}
void loop()
{
```

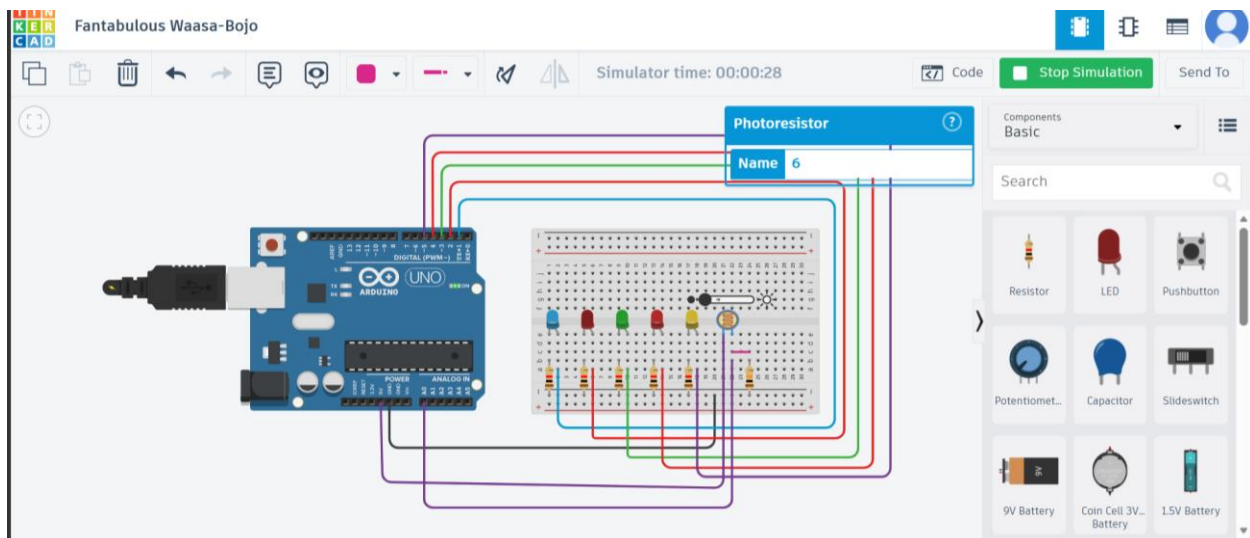
```

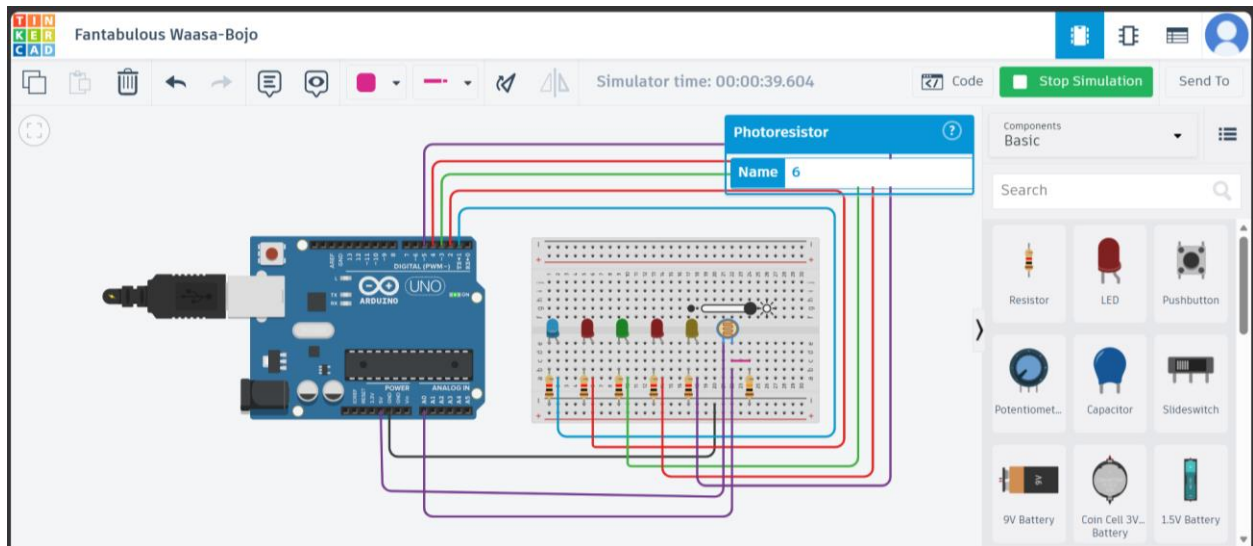
int light = analogRead(ldr);
Serial.println(light);
if(light < 500)    // Dark
{
    for(int i = 1; i <= 5; i++)
    {
        digitalWrite(i, HIGH);
        delay(200);
        digitalWrite(i, LOW);
    }
}
else                // Bright
{
    for(int i = 1; i <= 5; i++)
        digitalWrite(i, LOW);
}
}

```

5. Algorithm

1. Start the system.
2. Read the LDR value from analog pin A0.
3. Compare the value with the threshold.
4. If the value indicates darkness:
 - Turn on LEDs sequentially.
5. Otherwise:
 - Keep all LEDs OFF.
6. Repeat continuously.





6. Applications

- Smart street lighting
- Smart homes
- Garden lighting
- Parking lot illumination
- Campus lighting
- Energy-saving outdoor lighting

7. Advantages

- Automatic operation
- Reduces electricity consumption
- Low-cost implementation
- Easy to install
- Reliable and efficient
- Suitable for IoT-based smart city applications

8. Conclusion

This project demonstrates how an LDR sensor and Arduino can be used to automate lighting based on ambient light conditions. The simulation shows an effective and energy-efficient

solution for smart lighting systems, making it a simple example of sensor-based automation in IoT applications.